

Lab Report 8 – ATWOOD'S MACHINE

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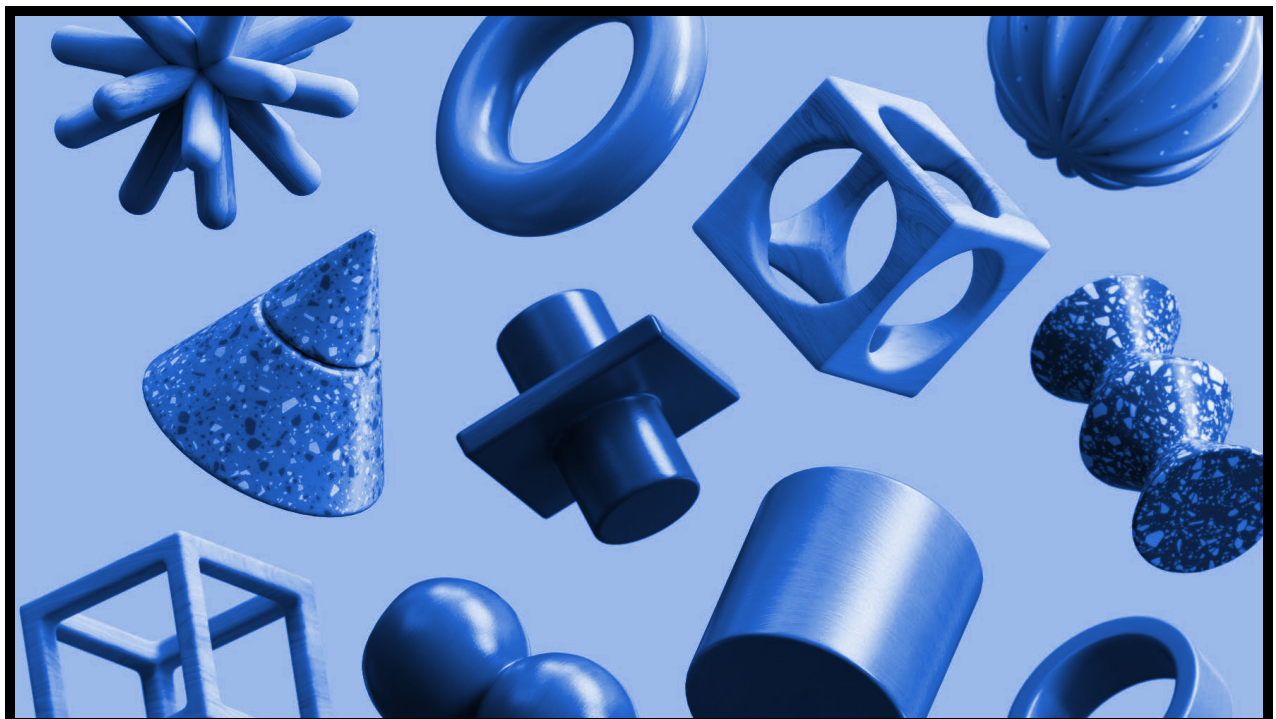
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Objectives

To confirm the theoretically predicted acceleration of an Atwood's machine.

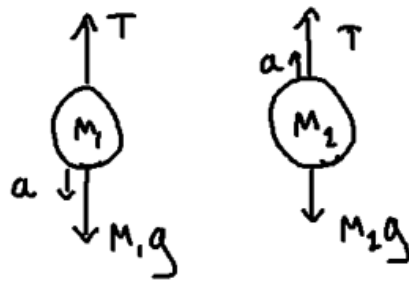
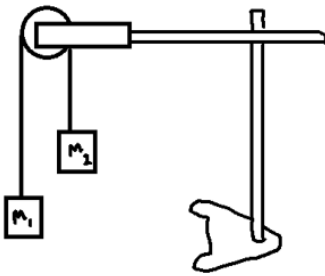
Equipments

The list of equipments used in this experiment is:

- Hanging masses
- Lab stand
- Photogate
- Pulley
- String
- Computer interface

Theory and Equations

An Atwood's machine consists of 2 masses hanging from the same pulley, as shown in the picture here.



Assume Mass 1 > Mass 2:

1. Motion Acceleration

$$F_{Net, 1, y} = M_1(-a) = T - m_1g$$

$$F_{Net, 2, y} = M_2(-a) = T - m_2g$$

$$\Rightarrow a = \frac{m_1 - m_2}{m_1 + m_2}g \quad (\text{Eq. 1})$$

2. Percentage Error

The percentage error is calculated based on the experimentally determined measure (exp) and the accepted value (acc). And there are 2 types of accepted values which are theoretical value which is predicted from theory and empirical value which is collected from a reliable source.

$$\text{Percentage Error} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% \quad \text{units: \%}$$

Summary of Procedures

To begin, we establish a stable experimental setup using the necessary equipment, as illustrated in the image above. The initial masses are set to 201.0 g for mass 1 and 199.0 g for mass 2. While recording in the computer interface, the masses are released from rest. The acceleration is determined from the velocity versus time graph by applying a linear fit, where the slope corresponds to the acceleration. We document the measured acceleration along with the respective mass values. Subsequently, 0.5g is transferred from mass 2 to mass 1, and the procedure is repeated to obtain a new acceleration value. This process continues iteratively until a total of 30 acceleration measurements are recorded. Additionally, a hanging mass of 5g is incorporated into the final mass of each object.

Data and Observations

Note:

- The masses in the table below are already added 5g from the hanging mass.
- m1 = mass 1
- m2 = mass 2
- exp a = experimental acceleration recorded by the computer interface

Id	m1 (g)	m2 (g)	exp a (m/s ²)
1	206.0	204.0	0.0430
2	206.5	203.5	0.0690
3	207.0	203.0	0.0920
4	207.5	202.5	0.117
5	208.0	202.0	0.138
6	208.5	201.5	0.164
7	209.0	201.0	0.186
8	209.5	200.5	0.206
9	210.0	200.0	0.230
10	210.5	199.5	0.255
11	211.0	199.0	0.277
12	211.5	198.5	0.303
13	212.0	198.0	0.325
14	212.5	197.5	0.325
15	213.0	197.0	0.368
16	213.5	196.5	0.393
17	214.0	196.0	0.419
18	214.5	195.5	0.448
19	215.0	195.0	0.470
20	215.5	194.5	0.489
21	216.0	194.0	0.519
22	216.5	193.5	0.540
23	217.0	193.0	0.564
24	217.5	192.5	0.583
25	218.0	192.0	0.607
26	218.5	191.5	0.632
27	219.0	191.0	0.653
28	219.5	190.5	0.680
29	220.0	190.0	0.701
30	220.5	189.5	0.724

Data Analysis

Id	m1 (g)	m2 (g)	exp a (m/s ²)	acc a (m/s ²)	Percentage Error (%)	acc Slope (m/kg.s ²)
1	206.0	204.0	0.0430	0.048	10.1	23.9
2	206.5	203.5	0.0690	0.072	3.78	23.9
3	207.0	203.0	0.0920	0.10	3.78	23.9
4	207.5	202.5	0.117	0.12	2.10	23.9
5	208.0	202.0	0.138	0.14	3.78	23.9
6	208.5	201.5	0.164	0.17	1.98	23.9
7	209.0	201.0	0.186	0.19	2.73	23.9
8	209.5	200.5	0.206	0.22	4.24	23.9
9	210.0	200.0	0.230	0.24	3.78	23.9
10	210.5	199.5	0.255	0.26	3.01	23.9
11	211.0	199.0	0.277	0.29	3.43	23.9
12	211.5	198.5	0.303	0.31	2.49	23.9
13	212.0	198.0	0.325	0.33	2.88	23.9
14	212.5	197.5	0.325	0.36	9.35	23.9
15	213.0	197.0	0.368	0.38	3.78	23.9
16	213.5	196.5	0.393	0.41	3.28	23.9
17	214.0	196.0	0.419	0.43	2.61	23.9
18	214.5	195.5	0.448	0.45	1.35	23.9
19	215.0	195.0	0.470	0.48	1.68	23.9
20	215.5	194.5	0.489	0.50	2.58	23.9
21	216.0	194.0	0.519	0.53	1.30	23.9
22	216.5	193.5	0.540	0.55	1.77	23.9
23	217.0	193.0	0.564	0.57	1.68	23.9
24	217.5	192.5	0.583	0.60	2.44	23.9
25	218.0	192.0	0.607	0.62	2.33	23.9
26	218.5	191.5	0.632	0.65	2.07	23.9
27	219.0	191.0	0.653	0.67	2.43	23.9
28	219.5	190.5	0.680	0.69	1.90	23.9
29	220.0	190.0	0.701	0.72	2.24	23.9
30	220.5	189.5	0.724	0.74	2.29	23.9

acc g (m/s ²)	9.800
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Sample Calculations:

Case 1: $m_1 = 206.0\text{g}$ and $m_2 = 204.0\text{g}$

Following to the Eq. 1, we have:

$$\text{Accepted Acceleration} = \text{acc } a = \frac{m_1 - m_2}{m_1 + m_2} g = \frac{206.0 - 204.0}{206.0 + 204.0} 9.800 = 0.048 \text{ m/s}^2$$

$$\text{Percentage Error} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% = \left| \frac{0.043 - 0.048}{0.048} \right| * 100\% = 10.1 \%$$

$$\text{Accepted Slope} = \text{acc Slope} = \frac{g}{m_1 + m_2} = \frac{9.800}{206.0 + 204.0} * 10^3 = 23.9 \text{ m/kg} \cdot \text{s}^2$$

Results

Sample Calculation:

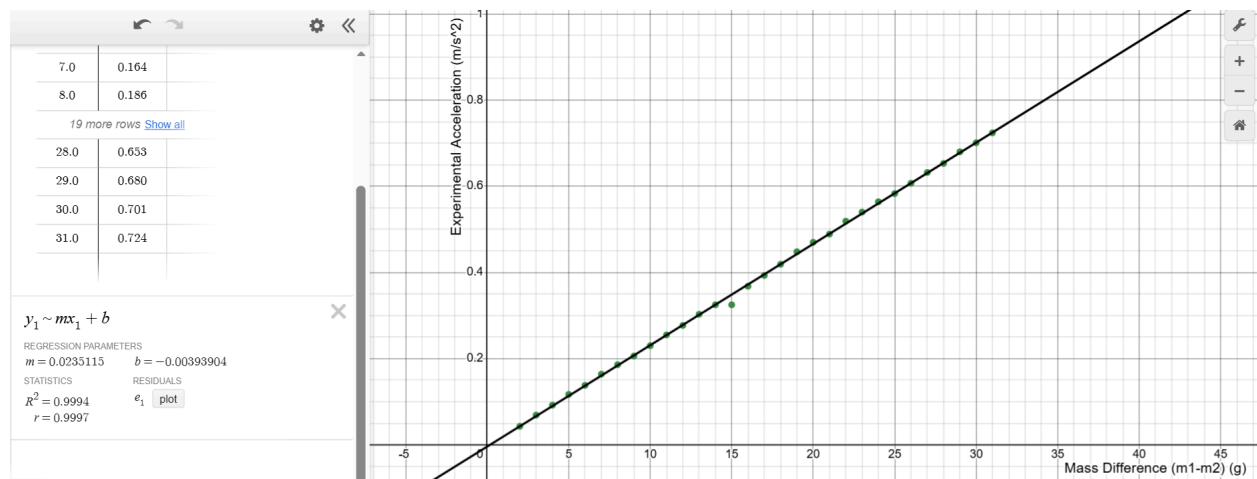
Case 1: $m_1 = 206.0\text{g}$ and $m_2 = 204.0\text{g}$. We have:

$$m_1 - m_2 = 206.0\text{g} - 204.0\text{g} = 2.00 \text{ g}$$

m1 - m2 (g)	exp a (m/s ²)
2.00	0.0430
3.00	0.0690
4.00	0.0920
5.00	0.117
6.00	0.138
7.00	0.164
8.00	0.186
9.00	0.206
10.0	0.230
11.0	0.255
12.0	0.277
13.0	0.303
14.0	0.325
15.0	0.325

16.0	0.368
17.0	0.393
18.0	0.419
19.0	0.448
20.0	0.470
21.0	0.489
22.0	0.519
23.0	0.540
24.0	0.564
25.0	0.583
26.0	0.607
27.0	0.632
28.0	0.653
29.0	0.680
30.0	0.701
31.0	0.724

Producing a graph of experimentally measured acceleration (vertical) versus $m_1 - m_2$ (horizontal), we have a graph shown as below.



From the graph, we got the value of experimental slope = exp slope = 0.0235115.

Applied the Percentage Error formula, we have:

$$\text{Percentage Error of Slope} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% = \left| \frac{0.0235115 - (23.9/1000)}{(23.9/1000)} \right| * 100\% = 1.64\%$$

Discussion / Conclusion

From the above graph of experimentally measured acceleration versus $(m_1 - m_2)$, we can see that as the mass difference increases, the acceleration also increases linearly. It shows clearly that the acceleration has a positive linear relationship with the mass difference $(m_1 - m_2)$. This confirms the theoretical acceleration in Eq. 1. For each record, 0.5g is transferred from mass 2 to mass 1, so the sum of $(m_1 + m_2)$ is constant (410 grams). We also have that the theoretical acceleration increases in the positive direction as the mass difference increases as shown below.

$$a = \frac{g}{m_1 + m_2} (m_1 - m_2) = \text{constant} * (m_1 - m_2) = \text{slope} * (m_1 - m_2)$$

Furthermore, with a very small percentage error of slope (1.64%), this shows that the experiment we performed to measure the acceleration is highly accurate. Again, this confirms that the ratio of acceleration to mass difference is equal to the slope, that is, the ratio of gravitational acceleration to total mass (graph slope $\sim \frac{g}{m_1 + m_2}$).

Post-Lab Question

The concept of Atwood's machine, which involves lifting a smaller mass using a larger one, can be observed in construction when workers manually hoist materials to higher levels before a staircase is completed. A more familiar real-world application is an elevator. This explains why elevators have weight limits, such as 450 kg for 6 people, since if the total weight inside the elevator (the smaller mass) closely matches the counterweight, the system may remain at rest. If the mass difference is too small, the elevator moves more slowly and consumes extra energy. This can also generate excess heat which can be detrimental to the service life of the elevator. By using counterweights of appropriate weight and setting the correct human weight limit, we can reduce the force required to operate the elevator, save energy and service life of the elevator. Furthermore, by adjusting the counterweight precisely, the elevator can bring people to the exact height of the house we want at a desirable speed.